

Beyond predictive benchmarks: infrastructure for hypothesis-driven cognitive testing of large language models

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Abstract

Large language models are increasingly evaluated as psychological subjects, but NeuroAI and psychology apply incompatible standards: predicting responses to naturalistic stimuli versus manipulating controlled stimuli to test falsifiable, mechanism-level hypotheses. No infrastructure exists for running the second kind of study at the scale quantitative psychology expects, so researchers hand-build prompt construction, memory management, and persona conditioning for every task and model. We introduce psychscanner, an open-source framework in which a researcher authors a cognitive or survey task once, as a declarative task card, and runs it across memory architectures, persona manipulations, feedback regimes, and model families without rewriting any mechanics. A three-armed bandit task run through three agent architectures found none produced explore-then-exploit behaviour, with instruction-following collapse under structured feedback as the dominant effect. A 480-trial factorial replication of a classic imagery paradigm found persona conditioning had no measurable effect on vividness ratings, but single-turn prompting returned an identical rating on every trial, a response-degeneracy artifact invisible to aggregate benchmark scores. Because every model backend implements one interface, the same task cards also capture internal activations: linear probes recover an internal board-state representation in a trained toy transformer, placing behavioural and mechanistic testing in one harness.

Keywords: cognitive testing, large language models, task-based evaluation, experimental psychology, reproducible evaluation

1 Introduction

Large language models are increasingly evaluated as objects of psychological inquiry [? ?], but the communities doing this evaluation disagree about what counts as evidence. NeuroAI typically evaluates a model by how well it predicts behavioural and neural responses to naturalistic stimuli, treating out-of-sample predictive variance as the mark of success [?]. Psychology instead evaluates a model by how well it accounts for the effect of systematically manipulating a controlled, artificial stimulus to test a specific, falsifiable hypothesis about a specific mechanism [?]. A model can predict naturalistic data well while getting the underlying mechanism wrong, and a model tuned to a controlled manipulation may generalize poorly outside it — the two standards are not interchangeable, and recent work has framed this as a live, unresolved methodological disagreement rather than a settled question: a 2026 Cognitive Computational Neuroscience Generative Adversarial Collaboration sets researchers who evaluate models by predictive fit to naturalistic stimuli against researchers who evaluate models by their account of controlled, manipulation-driven effects, and finds no consensus resolution [?]. The disagreement is concrete, not abstract: a 2026 preprint argues that Centaur [?] — the very foundation model just cited for NeuroAI’s predictive successes — mistakes predictive power for explanatory power [?], and even its critics hold that manipulation-based testing should complement rather than replace predictive modelling [?], the complement this paper builds for large language models.

Applying psychology’s side of this methodology to large language models at scale — systematically manipulating one factor (memory, persona, feedback, stimulus modality) across many models and replications, the way a cognitive psychology lab would run a study — currently has no dedicated infrastructure: prompt construction, response parsing, state management, and persona conditioning are typically hand-built from scratch for every new task and model family, which makes systematic, multi-condition replication impractical. We introduce **psychscanner**, an open-source framework that closes this gap: a researcher authors a controlled cognitive or survey task once, as a declarative task card, and runs it across memory conditions, persona manipulations, feedback regimes, and model families without rewriting the underlying mechanics — the missing methodological toolkit for asking manipulation-based, falsifiable questions about large language model cognition at the scale quantitative psychology expects.

Existing behavioural instruments for this purpose — benchmarks such as PsychoBench [?], task ontologies such as the Cognitive Atlas [?], and the methodological guidance now being formalized for treating models as psychological simulators [?] — give the enterprise structure and coverage targets, but share one limit: they describe what a model does without opening it to ask why. That limit is consequential, because the behavioural divergences this literature keeps surfacing are exactly the ones whose cause is unclear. Two studies run on earlier versions of the framework reported here illustrate the pattern: model source-monitoring accuracy degrades under conversational delay in a way that tracks active rather than aggregate parameter count [?], and model imagination-vividness ratings collapse into a degenerate single-cluster network topology where three geographically distinct human populations show robust community structure [?]. Both findings describe behaviour and stop there.

We therefore designed psychscanner so that the layer which chooses *which model executes a trial* is a single, swappable interface, and then implemented backends behind that interface that additionally capture the model’s internal activations for every trial. This is what makes the framework more than a prompting harness: the three dominant mechanistic-interpretability methods have direct neuroscience analogues — linear probing of internal activations is the same operation as neural decoding from recorded population activity [?]; causal tracing, in which an activation is corrupted and the site that restores correct behaviour is identified, is the artificial-network analogue of a lesion or reversible-inactivation study [?]; and activation steering along an extracted direction is the analogue of stimulation along an identified pathway [? ?]. Because these run through the same task cards, persona conditioning, and memory conditions used for the behavioural experiments, a manipulation that previously yielded only response differences now also yields paired activation captures, with no separate extraction pipeline to write. We treat internal activations as a complement to behavioural measurement rather than a replacement for it: mechanistic interpretability’s own recent self-assessment favours pragmatic proxy tasks over ambitious reverse-engineering [?], and a clinical-triage study found that near-perfect internal representations (98.2% AUROC from linear probes) did not translate into corrected model output (steering fixed only 20% of missed errors) [?].

2 Results

2.1 Framework architecture

Figure ?? gives the component structure of the released package (v0.3.0). A run traverses a fixed spine — the researcher-authored task card and persona files are resolved into an **ExpCard**, expanded into one system message per design cell and one message per trial, executed by **TaskRunner** against a configured agent, and validated and written to disk — while four seams remain open for the researcher: response parsers, model providers, agent architectures, and trial-level feedback. The experimental factors a psychologist would want to manipulate are all declarations on the card or the **ExpCard** rather than code paths, and the model provider seam is what carries the interpretability backends (§??). Six demonstration modules exercise this feature surface end-to-end on real model calls; the two with complete behavioural data are reported below, alongside the interpretability validation.

2.2 Behavioural manipulations

A 3-armed bandit task run through three agent architectures (a bare single-call baseline, a generate/reflect loop, and a plan/execute/validate loop) via a single `custom_agent=` argument found none produced textbook explore-then-exploit behaviour (baseline: 16.14 total reward, 0% optimal-arm picks; self-reflection: 14.14, 16.7% optimal; plan/execute/validate: 12.14, 0% optimal; $N=18$ rounds, demonstration-scale, not a powered study); the dominant effect was instruction-following collapse under structured JSON feedback on an 8B-parameter model, not a clean architecture comparison (Fig. ??, Table ??).

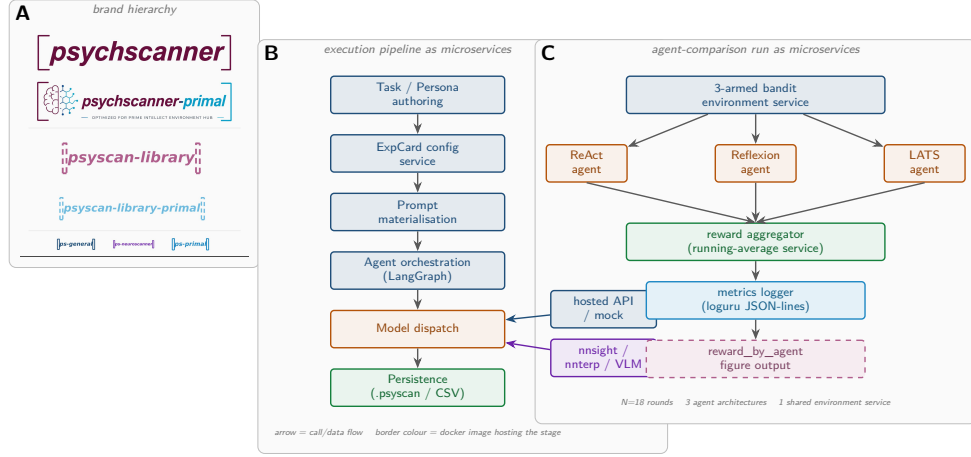


Fig. 1 The PsychScanner project as a microservices system. (A) Brand hierarchy: the two framework repositories (psychscanner, psychscanner-primal), their companion vetted task/experiment-card libraries (psyscan-library, psyscan-library-primal), and the three published Docker images (ps-general, ps-neuroscanner, ps-primal). (B) The single-run execution pipeline of §??, redrawn as a service pipeline; box border colour marks which Docker image hosts each stage. (C) The multi-agent bandit comparison behind Fig. ??, redrawn the same way: one shared environment service, three interchangeable agent services, and a reward-aggregation/logging path terminating in that figure.

Table 1 Demo 01 per-agent summary. $N = 18$ rounds total (6 rounds $\times$ 3 architectures), demonstration-scale, not a powered study.

Agent architecture	Total reward	Mean reward/round	% optimal-arm picks
Baseline (single-call)	16.14	2.69	0%
Self-reflection	14.14	2.36	16.7%
Plan/execute/validate	12.14	2.02	0%

[PLACEHOLDER: paired-associate recall across memory/feedback conditions; BFI-44 persona $\times$ memory survey; vision-language accuracy under reward/accuracy feedback; and persona-vector extraction and steering — design complete (see Methods), quantitative results pending full data collection.] A completed extended factorial replication of the persona/imagery paradigm (VVIQ-16, Supplementary Information S1.1, $n = 480$ trials) found no measurable effect of persona conditioning on vividness ratings (WEIRD 3.69 ± 0.88 vs. non-WEIRD 3.70 ± 0.89), but a strong memory-condition effect: the persona-conditioned single-turn condition returned a constant rating of 4 on all 160 trials ($SD = 0$), whereas the trial-chain condition’s ratings spanned the full 1–5 range (3.02 ± 1.28 ; Fig. ??, Table ??). Further extended factorial designs beyond the six core demos are described in Supplementary Information Sections S1–S4.

Table 2 VVIQ-16 mean vividness rating ($\pm$ SD) by memory condition, pooled across personas ($n = 160$ trials/condition). SD = 0 under `item.ST` indicates every trial returned an identical rating.

Memory condition	Mean vividness	SD
<code>item.ST</code> (single-turn)	4.00	0.00
<code>task.Cv</code> (episodic-chain)	4.06	0.24
<code>trial.Cv</code> (trial-chain)	3.02	1.28

2.3 Activation capture through the same interface

To test whether the model-provider seam in Fig. ?? really does carry mechanistic work without a parallel pipeline, we reproduced the Othello-GPT result [?] end-to-end through it. A CPU-tractable toy transformer (4 layers, $d_{\text{model}} = 64$, 4 heads) was trained for 12 epochs on 4,000 games of legal-move sequences only, with board state never shown during training, and then linearly probed for board-state decodability at each residual-stream layer on 400 held-out games (Methods, §??). Legality checking on the trained model (728 boundary steps) placed a mean probability mass of 0.998 on legal moves against a random baseline of 0.75, confirming the model had learned the task before its internals were probed.

Board-state decodability is near the shuffled-label baseline at layer 0 (0.423 vs. 0.140) and jumps above 0.93 from layer 1 onward (Table ??), the same qualitative layer-wise pattern the original study reported, obtained here on a model we trained ourselves rather than a reused checkpoint, through the infrastructure built for the behavioural experiments above.

Table 3 Linear-probe accuracy for an internal board-state representation by residual-stream layer, in a toy transformer trained only on legal-move sequences. The shuffled-label baseline is computed identically at each layer, to control for probe capacity alone recovering structure.

Layer	Probe accuracy	Shuffled-label baseline	Above baseline
0	0.423	0.140	+0.283
1	0.941	0.148	+0.793
2	0.953	0.149	+0.804
3	0.931	0.141	+0.790

[PLACEHOLDER: the two remaining planned reproductions — causal tracing and rank-one editing [?], and contrast-consistent search [?] — are implemented as designs but have not been run, and the persona-vector extraction and steering experiment [?] has no collected data as of this draft. The claim that this backend is validated as a general-purpose probing platform is therefore currently supported by one of three planned reproductions.]

3 Discussion

The VVIQ-16 result is the clearest evidence so far for the methodological claim in the Introduction. A naturalistic or aggregate-benchmark evaluation of this model would report a mean vividness rating around 3.7 and move on — a plausible-looking number, indistinguishable from genuine item-level imagery report unless someone looks inside the condition. Decomposing by memory architecture is what exposes that the persona-conditioned single-turn condition was not discriminating between items at all: every one of 160 trials returned the identical rating, a response-degeneracy artifact rather than a vividness judgment. A persona-free replication of the same task and model does not show this complete collapse under single-turn prompting — ratings there are concentrated but not degenerate ($M = 3.81$, $SD = 0.66$), broadening under conversational memory ($M = 3.73$, $SD = 0.99$) [?]. The two results together implicate an interaction rather than either factor alone: persona conditioning narrows the response range, and adding a no-memory constraint on top of it collapses that range entirely, in a way memory manipulation by itself does not. That interaction is exactly the kind of finding a one-factor-at-a-time evaluation cannot surface, and precisely why varying persona and memory as independent, crossable factors — rather than as incidental properties of whichever setup happened to be convenient — matters. Prediction-on-naturalistic-stimuli evaluation, of the kind foundation-model benchmarking typically performs [?], has no mechanism for catching this, because nothing in that evaluation paradigm asks what happens when one experimental factor is held fixed while another is varied. That is precisely the manipulation-based standard psychology applies to modelling claims [?], and precisely what a task card makes practical to run at scale.

The bandit result generalizes the same lesson to a different failure mode. None of the three agent architectures reached the optimal arm reliably, but the more informative finding was that all three, not just one, stopped emitting the requested single-letter choice once structured JSON feedback entered the prompt — an instruction-following collapse that a per-architecture reward comparison would have silently absorbed into "agent B was worse" rather than surfacing as a property of the feedback format itself. In both cases here, the informative result was not the headline metric but what varying one factor at a time revealed underneath it.

Both of those findings, however, are of the same kind as the ones that motivated this work: they say what the model did, not why. The board-state probe (Table ??) is what shows the framework can in principle answer the second question with the same task cards, and it is worth being precise about how far that goes. What the reproduction establishes is infrastructural: activations captured through the ordinary model-provider interface, on a model we trained ourselves, support a probe that recovers the layer-wise pattern the original Othello-GPT study reported [?] — early layers carrying surface features of the move sequence, task-structured internal state becoming linearly accessible only once the network has had a layer to compose it. What it does not establish is anything about the divergences in the Introduction. Neither the reality-monitoring degradation [?] nor the imagination-network collapse [?] has been probed mechanistically here. That is a gap rather than a hedge, and it names the next experiment precisely: run the same reality-monitoring and VVIQ-16 task cards through the activation-capturing backend and ask whether the single-turn response

degeneracy reported above corresponds to reduced representational variance in the residual stream, or is a decoding-time artifact with no mechanistic signature at all. The framework makes that a configuration change rather than a new study.

Limitations. Both completed behavioural results are demonstration-scale on small models (16–18 rounds; 480 trials on a 360M-parameter model), not powered studies, and only two of the six planned demonstration modules have analyzed behavioural data as of this draft. The interpretability results rest on one probe run per layer of a single toy transformer, and on one of three planned reproductions: neither causal tracing [?] nor contrast-consistent search [?] has been run. Whether the single-turn response-degeneracy pattern is a small-model artifact or persists at larger scale — and whether it disappears once memory or model capacity crosses some threshold — is the direct next test the framework already supports: rerun the same task card across a range of model sizes with model family and memory architecture as the two manipulated factors, and ask at what capacity, if any, single-turn item discrimination recovers.

4 Methods

4.1 Task cards

A *task card* is the declarative, hand-authorable unit a researcher writes to define an experiment: a JSON document (or equivalent inline Python dict) specifying instructions, a set of named contexts, and the trials themselves, with no code required for the common case. Each trial carries a unique identifier (**trcode**) and a stimulus (plain text, a structured dict, or a list of multimodal content blocks); the text before the first underscore in a trial’s **trcode** is matched against the card’s **contexts_id** list to derive a **context_item** — the full-text description of the condition that trial belongs to — which is attached automatically to every trial for prompting and for output records, so authors write the mapping once rather than repeating context text per trial.

A single **chain_type** field controls how trials share conversational state, independent of which specific memory mode (stateless **SingleTurn** or continuous **Convo**) is active: **item** sends each stimulus as an independent call; **trial** groups multiple stimuli under one **trcode** into a single conversational thread (e.g. a study phase followed by its matching test phase); **task** places every trial in the run on one continuous thread. This single parameter is what lets the same underlying trial-execution engine implement single-turn, trial-chain, and episodic-chain designs (Demos 2 and 3; Supplementary Information Sections S1–S2) without separate code paths. Structured response parsing (a Pydantic schema per trial or per card), per-trial tool subsets, and trial-level feedback injection are declared the same way, as optional fields on the same card.

4.2 Persona conditioning and memory conditions

Persona conditioning is authored as one or more JSON files, each holding a list of persona statements. At prompt-build time the framework takes the Cartesian product across files and emits one system message per resulting cell, so a two-file design with

two levels each yields four conditions without the researcher enumerating them. Memory is configured independently: **SingleTurn** compiles the execution graph without a conversation checkpoint, and **Convo** compiles it with one. Windowing and summarization are orthogonal parameters rather than separate modes — **memory_k** bounds how much history is retained, and **summary_k** folds the overflow into a rolling summary prepended to the system message — which is what makes “stateless versus windowed versus summarized memory” a factor that can be crossed with persona and feedback in one design rather than three implementations.

4.3 Model providers and activation capture

Every model backend is reached through a single dispatch function and implements the same **BaseChatModel** interface, so the choice of model is a field on the **ExpCard** rather than a code path (Fig. ??). Hosted APIs, a deterministic mock model used for offline testing, and three local backends built on **nnsight** are all interchangeable at that seam. The local backends (**ChatNNSightModel**, **ChatNNterpModel**, and a vision-language variant) additionally run a cached forward pass over each prompt and write the captured per-module residual-stream activations to disk, referencing the file from the returned message rather than embedding tensors in it — the same externalization the framework applies to multimodal stimuli. Two consequences follow. Persona conditioning, memory modes, task cards, and feedback hooks apply unchanged to an activation-capturing run; and capture is over the prompt’s forward pass, not per generated token, which bounds what can be asked of the resulting tensors.

For the reproduction reported in §?? we trained a 4-layer transformer ($d_{\text{model}} = 64$, 4 attention heads) for 12 epochs on 4,000 games represented purely as legal-move sequences; board state was never part of the training signal. Board-state decodability was then measured by fitting a linear probe to the residual stream at each layer, on 400 held-out games. The reported baseline is a probe of identical capacity fitted at the same layer to shuffled labels, which is what distinguishes a recovered representation from the probe itself imposing structure. Model competence was verified separately by measuring the probability mass the trained model placed on legal moves at 728 boundary steps.

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Declarations

- Funding: PLACEHOLDER
- Conflict of interest/Competing interests: PLACEHOLDER
- Ethics approval: Not applicable
- Consent to participate: Not applicable
- Consent for publication: Not applicable
- Availability of data and materials: PLACEHOLDER
- Supplementary information: This article has accompanying Supplementary Information (**supplementary.tex**), describing extended factorial-design experiments (Sections S1–S4) beyond the six core demonstration modules in the main text.

- Code availability: Public repository at <https://github.com/saurabhr/psychscanner>, tagged release to be added once the version used for this submission's results is frozen.
- Authors' contributions: PLACEHOLDER

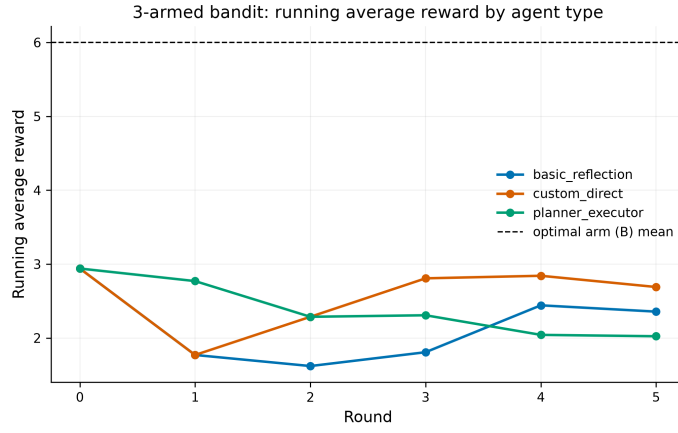


Fig. 3 Running average reward by agent architecture across the 3-armed bandit task (Demo 01). The optimal arm’s true mean (6.0) is shown as a dashed reference line; none of the three architectures approaches it.

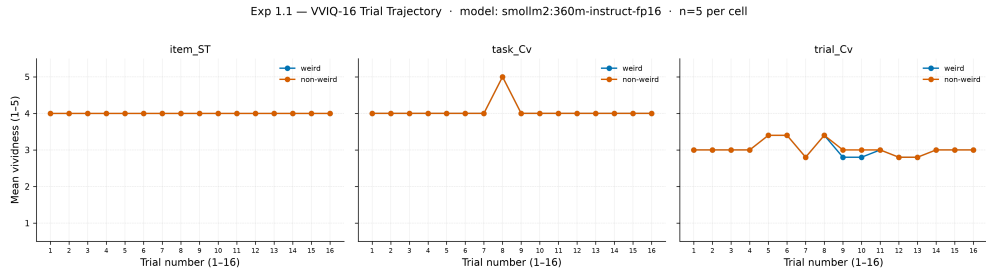


Fig. 4 VVIQ-16 vividness ratings by trial, memory condition, and persona (Supplementary Information S1.1). `item_ST` (single-turn) is flat at rating 4 for every trial in both personas; `trial_Cv` shows genuine item-to-item variation.